Measurement Fundamentals

Week 5 — Tuesday Lecture

Today

- Accuracy, precision, and resolution — three different things
- Where measurement error actually comes from
- How much accuracy a decision needs
- Reporting a number you can defend

The question this course asks

"The lot is 210 feet long" — an assertion.

"210 ± 6 feet" — a measurement.

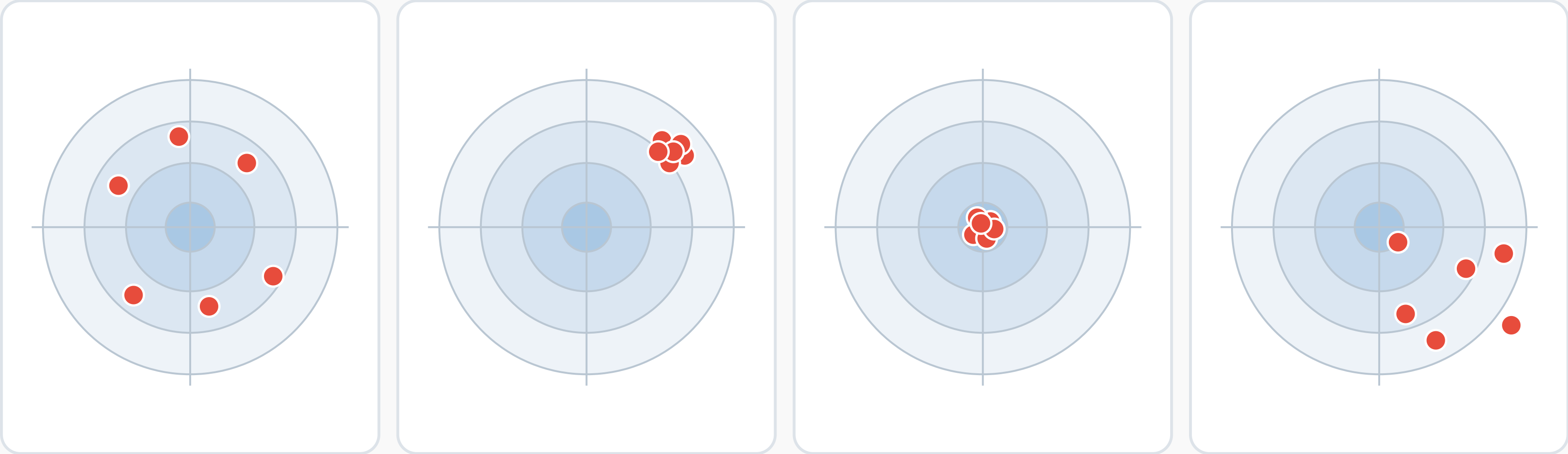
Every method in this course, from pacing to a drone, is judged by the same test.

How good is this number?

You will not know until you can answer that with a range, not a single value.

Accuracy and precision are two different things

The bullseye is the true value. Each dot is one measurement of the same thing.



Accurate, not precise
Right on average, badly scattered

Precise, not accurate
Repeatable, and repeatedly wrong

Accurate and precise
What you are aiming for

Neither
Scattered, and off as well

Averaging fixes scatter. It does not fix an error that pushes every measurement the same way.
That is why you need both numbers, and why one of them cannot stand in for the other.

Accurate means close to the true value. Precise means the repeats agree.

Three different things

- **Accuracy** — how close to the true value. Errors are usually **systematic**.
- **Precision** — how closely repeats agree with each other. Errors are usually **random**.
- **Resolution** — the smallest difference the instrument can show you.

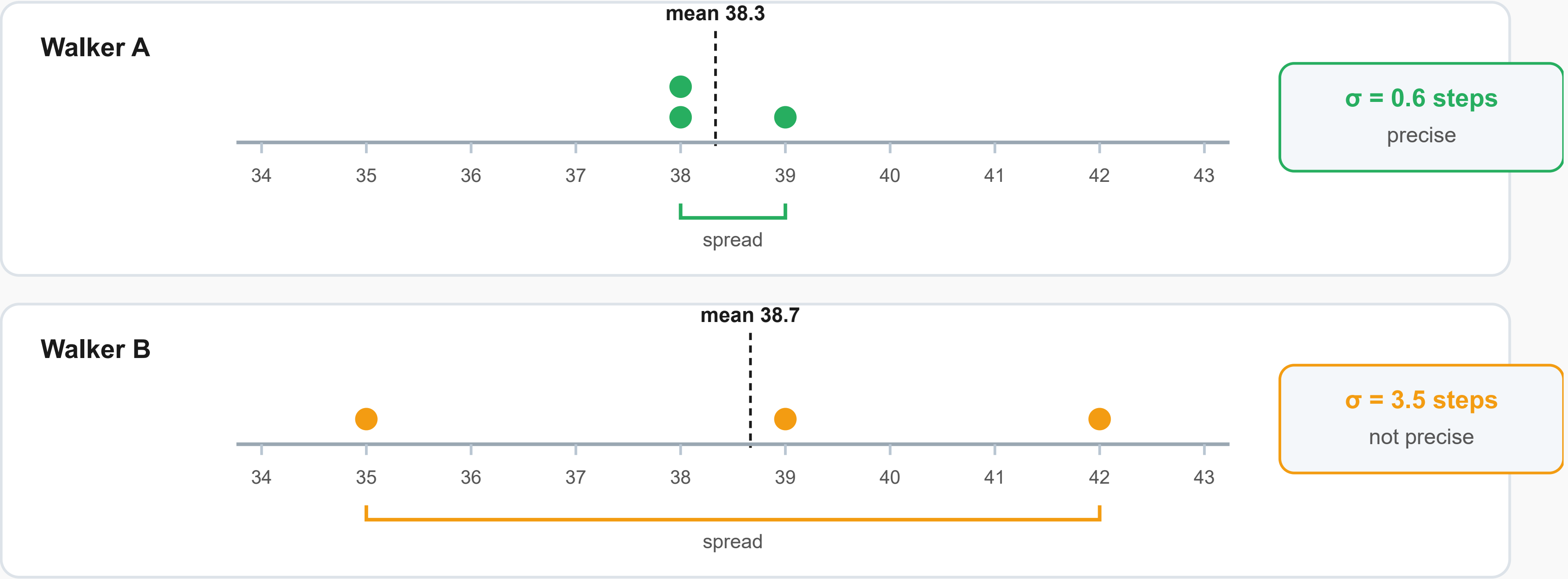
Averaging fixes scatter. It does not fix a result that is consistently off.

Which walker would you trust for a single walk?

Walker A: 38, 39, 38 steps   Walker B: 35, 42, 39 steps

Precision is the spread, and it has a number

Two students pace the same 100 ft three times. Their step counts:



Both walkers average about the same count. Only one of them can tell you how far to trust a single walk.

This figure is only about precision. Accuracy needs the tape.

Same average, very different spread — only one walker is precise.

Walker A's standard deviation

$$\bar{x} = \frac{38 + 39 + 38}{3} = 38.33$$

$$\sigma = \sqrt{\frac{(-0.33)^2 + (0.67)^2 + (-0.33)^2}{3 - 1}} = \sqrt{0.33} \approx 0.6 \text{ steps}$$

A tight σ on an average near 38 — Walker A is precise. This says nothing yet about accuracy.

Four places the error comes from

Only one of them is the instrument's fault.

The instrument

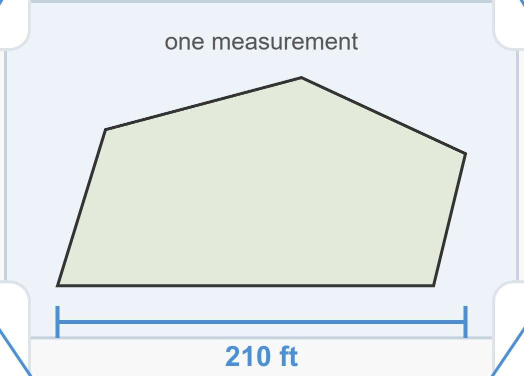
A tape that has stretched, or a scale too coarse

→ Check it against a known length

The method

Treating a lot that is not rectangular as a rectangle

→ Match the method to the shape



The operator

Stretching your stride while calibrating a pace

→ Repeat the measurement, then average

The reference

A benchmark that moved, or imagery three years old

→ Use a reference you can verify

Repeating a measurement exposes the operator. It hides the other three.

Instrument, method, operator, reference — repeating a measurement only exposes one.

Turning a percent into a range

A calibrated pace is good to about **3%**.

$$3\% \times 210\ \mathrm{ft} \approx 6\ \mathrm{ft}$$

Report **210 ± 6 ft** — not a bare 210.

Boulder, or beam?

"Is there a boulder in this field?" and "has this beam settled?" do not need the same accuracy.

The decision sets the accuracy, not the equipment

Work down the list only as far as the question actually requires.

MORE COST

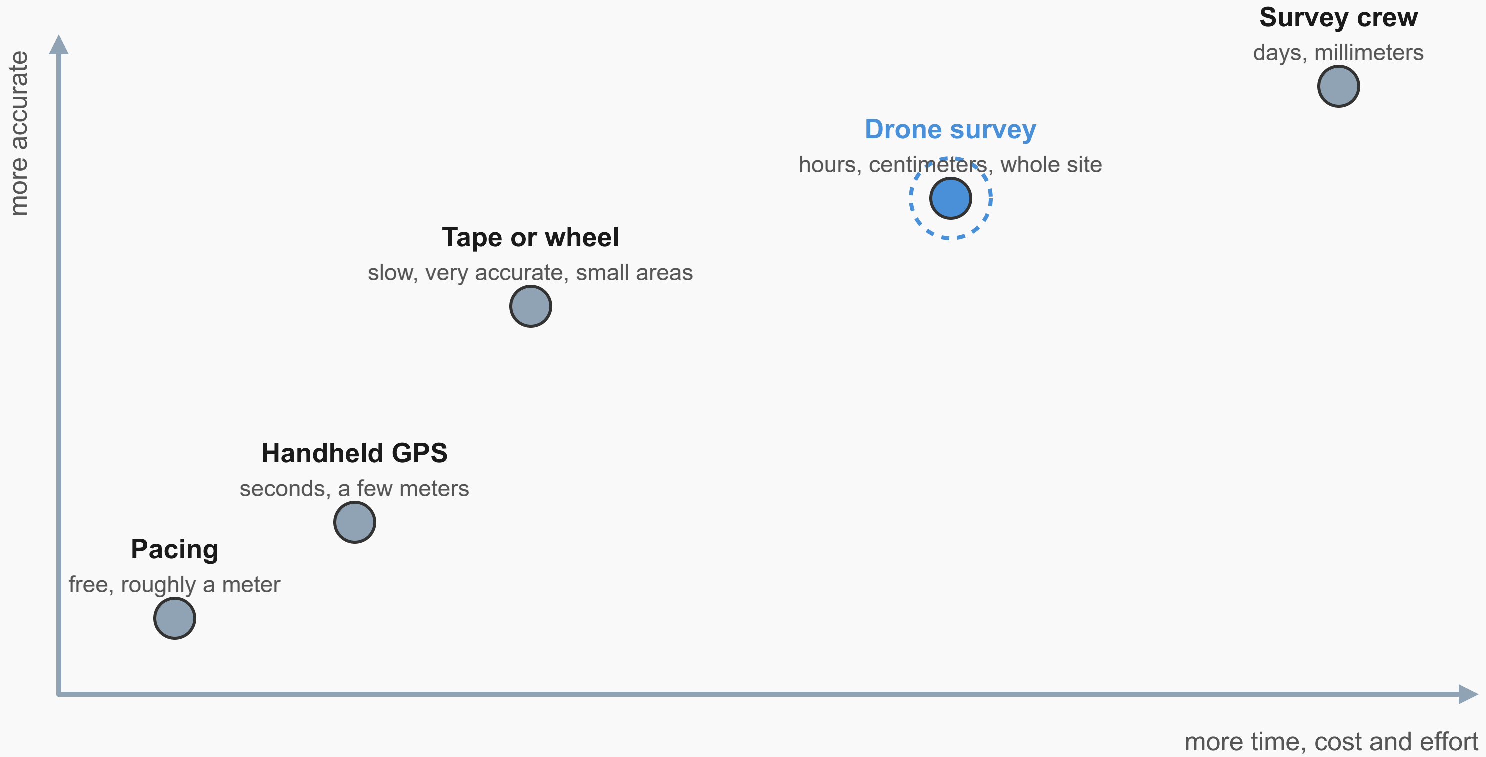
meters	Is there a boulder in this field?	pacing, or free imagery
about 1 m	Roughly how much asphalt does this lot need?	free imagery, or a wheel
10 cm	How much fill was moved this week?	aerial survey with ground control
1 cm	Does this pipe fall the whole way to the outlet?	level, or survey-grade GNSS
1 mm	Has this beam settled since last month?	total station, or dial gauges

Accuracy you do not need is wasted money. Accuracy you do need and did not buy is a failure you will not find out about until it matters.

The decision sets the accuracy needed — meters down to millimeters.

Picking a measurement method

There is no best method, only the cheapest one that still answers the question.



This course is about knowing which dot to reach for, and being able to explain why.

Accuracy rises with effort, but not smoothly — pick the method the decision needs.

Pace it off, or fly it?

A pair of boots and a drone both answer some question correctly, and both answer some question wastefully.

Your calculator does not know how good your measurement is

42 paces at 2.53 ft per pace, with a pace good to about three percent.

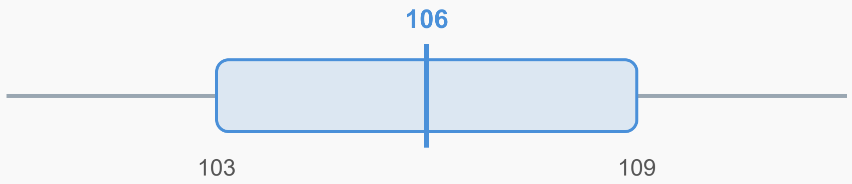
what the calculator says

106.~~26~~

decoration

what you report

106 ± 3 ft



three percent of 106 ft is about 3 ft, so the answer lives anywhere in here

Quote no more digits than your uncertainty supports, and put the uncertainty next to the number.
Extra digits do not make a measurement look careful. They make it look unexamined.

Your calculator does not know how good your input was.

106.26, or 106 ± 3?

42 paces at a calibrated 2.53 ft/pace:

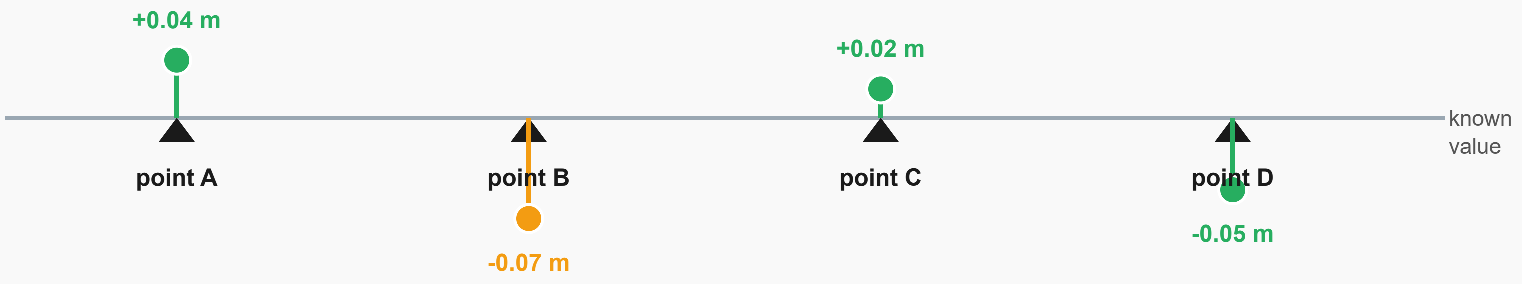
$$42 \times 2.53\ \mathrm{ft} = 106.26\ \mathrm{ft}$$

Pace is good to about 3% $\Rightarrow$ about ±3 ft.

Report 106 ± 3 ft. The extra digits are decoration, not confidence.

A check is only worth anything if it is independent

Four known points, measured again by your method. The gap is what you report.



What this tells you

Largest gap is 0.07 m. That is the honest claim,
not the finest detail you can see.

Independent means independent

Measuring the same way twice tests precision.
Only a different source tests accuracy.

When someone asks whether your measurement is any good, this is the answer you give them.

The largest residual against an independent check — that is the honest claim.

What to do Thursday

- Pace your calibration distance **three times** — record every count
- Compute your own $\bar{x}$ and σ before lab
- Bring both to Measurements and Methods

You will check your pace against a tape, then against a parking lot.